1、设 A 是可对角化 n 阶矩阵，其特征值为 $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots \geq |\lambda_n|$，相应特征向量为 $x^{(1)}, \cdots, x^{(n)}$。取向量 $v^{(0)} \in \mathbb{C}^n$, 假设 $v^{(0)} \notin \text{span}\{\text{x}^{(2)}, \cdots, \text{x}^{(\text{n})}\}$, 证明以下迭代过程

$$v^{(k+1)} = \frac{Av^{(k)}}{\|Av^{(k)}\|_2}, \quad k = 0, 1, \cdots$$

的相应 Rayleigh 商

$$R_k = \frac{\left(Av^{(k)}, v^{(k)}\right)}{\|v^{(k)}\|_2^2}, \quad k = 0, 1, \cdots$$

满足估计式

$$|R_k - \lambda_1| \leq Cr^k, \quad k = 0, 1, \cdots$$

其中 $C > 0$ 为一常数，$r \equiv |\frac{\lambda_2}{\lambda_1}|$.

证明：我们有$Ax^{(j)} = \lambda x^{(j)}$。我们将$v^{(k)}$表示为$\sum_{j=1}^{n} c_j^{(k)} x^{(j)}, c_1^{(0)} \neq 0$，那么有

$$v^{(k+1)} = \sum_{j=1}^{n} \frac{\lambda_j c_j^{(k)}}{\|Av^{(k)}\|_2^2} x^{(j)},$$

$$v^{(k)} = \left(\prod_{i=0}^{k-1} \frac{1}{\|Av^{(i)}\|_2^2}\right) \sum_{j=1}^{n} \lambda_j^k c_j^{(0)} x^{(j)}, c_j^{(k)} = \left(\prod_{i=0}^{k-1} \frac{1}{\|Av^{(i)}\|_2^2}\right) \lambda_j^k c_j^{(0)}.$$

根据题目条件，λ_1是唯一的模最大的特征值，因此有

$$\lim_{k\to\infty} \frac{c_j^{(k)}}{c_1^{(k)}} = \lim_{k\to\infty} \frac{\lambda_j^k}{\lambda_1^k} = 0, j = 1,2, \ldots, n.$$

我们考虑 Rayleigh 商

$$R_k = \frac{\left(Av^{(k)}, v^{(k)}\right)}{\|v^{(k)}\|_2^2} = \frac{\left(\prod_{i=0}^{k-1} \frac{1}{\|Av^{(i)}\|_2^2}\right)^2 \left(\sum_{j=1}^{n} \lambda_j^{k+1} c_j^{(0)} x^{(j)}, \sum_{j=1}^{n} \lambda_j^k c_j^{(0)} x^{(j)}\right)}{\left(\prod_{i=0}^{k-1} \frac{1}{\|Av^{(i)}\|_2^2}\right)^2 \left(\sum_{j=1}^{n} \lambda_j^k c_j^{(0)} x^{(j)}, \sum_{j=1}^{n} \lambda_j^k c_j^{(0)} x^{(j)}\right)}$$

$$= \frac{\sum_{j,l=1}^{n} \lambda_j^{k+1} \bar{\lambda}_l^k c_j^{(0)} \bar{c}_l^{(0)} \left(x^{(j)}, x^{(l)}\right)}{\sum_{j,l=1}^{n} \bar{\lambda}_j^k \bar{\lambda}_l^k c_j^{(0)} \bar{c}_l^{(0)} (x^{(j)}, x^{(l)})}.$$

$$R_k - \lambda_1 = \frac{\sum_{j,l=1}^{n} (\lambda_j - \lambda_1) \lambda_j^k \bar{\lambda}_l^k c_j^{(0)} \bar{c}_l^{(0)} \left(x^{(j)}, x^{(l)}\right)}{\sum_{j,l=1}^{n} \lambda_j^k \bar{\lambda}_l^k c_j^{(0)} \bar{c}_l^{(0)} (x^{(j)}, x^{(l)})}$$

$$= \frac{\sum_{j,l=1}^{n} (\lambda_j - \lambda_1) \left(\frac{\lambda_j}{\lambda_1}\right)^k \overline{\left(\frac{\lambda_l}{\lambda_1}\right)^k} c_j^{(0)} \bar{c}_l^{(0)} \left(x^{(j)}, x^{(l)}\right)}{\sum_{j,l=1}^{n} \left(\frac{\lambda_j}{\lambda_1}\right)^k \overline{\left(\frac{\lambda_l}{\lambda_1}\right)^k} c_j^{(0)} \bar{c}_l^{(0)} (x^{(j)}, x^{(l)})}$$

$$= \frac{\sum_{2\leq j\leq n, 1\leq l\leq n} (\lambda_j - \lambda_1) \left(\frac{\lambda_j}{\lambda_1}\right)^k \overline{\left(\frac{\lambda_l}{\lambda_1}\right)^k} c_j^{(0)} \bar{c}_l^{(0)} \left(x^{(j)}, x^{(l)}\right)}{\sum_{j,l=1}^{n} \left(\frac{\lambda_j}{\lambda_1}\right)^k \overline{\left(\frac{\lambda_l}{\lambda_1}\right)^k} c_j^{(0)} \bar{c}_l^{(0)} (x^{(j)}, x^{(l)})}.$$

$$(R_k-\lambda_1)\left(\frac{\lambda_1}{\lambda_2}\right)^k=\frac{\sum_{2\le j\le n,1\le l\le n}(\lambda_j-\lambda_1)\left(\frac{\lambda_j}{\lambda_2}\right)^k\overline{\left(\frac{\lambda_l}{\lambda_1}\right)^k}c_j^{(0)}\bar{c}_l^{(0)}(x^{(j)},x^{(l)})}{\sum_{j,l=1}^{n}\left(\frac{\lambda_j}{\lambda_1}\right)^k\overline{\left(\frac{\lambda_l}{\lambda_1}\right)^k}c_j^{(0)}\bar{c}_l^{(0)}(x^{(j)},x^{(l)})}.$$

对于上式的分子的模长，可以做如下的估计：

$$\left|\sum_{2\le j\le n,1\le l\le n}(\lambda_j-\lambda_1)\left(\frac{\lambda_j}{\lambda_2}\frac{\bar{\lambda}_l}{\bar{\lambda}_1}\right)^k c_j^{(0)}\bar{c}_l^{(0)}(x^{(j)},x^{(l)})\right|\le\left|\sum_{2\le j\le n,1\le l\le n}(\lambda_j-\lambda_1)c_j^{(0)}\bar{c}_l^{(0)}(x^{(j)},x^{(l)})\right|.$$

这是一个与k无关的**常数上界**。而当$k\to+\infty$时，分母的极限为$c_1^{(0)}\bar{c}_l^{(0)}(x^{(1)},x^{(1)})$这是一个正实数。这说明，$\left|(R_k-\lambda_1)\left(\frac{\lambda_1}{\lambda_2}\right)^k\right|$是有上界的。也即存在$C$使得

$$|(R_k-\lambda_1)|\le C\left|\frac{\lambda_2}{\lambda_1}\right|^k.$$

2、设 $b=\begin{pmatrix}1\\1\\1\end{pmatrix}$，

(1) 求一个 Householder 矩阵 H，使得 $Hb=\begin{pmatrix}c\\0\\0\end{pmatrix}$；

(2) 求若干个 Givens 矩阵之积 J，使得 $Jb=\begin{pmatrix}c\\0\\0\end{pmatrix}$.

备注：答案不是唯一的。

定义 5.4 (Householder 变换)

对 $\mathbf{v}\in\mathbb{C}^n$, $\mathbf{v}^*\mathbf{v}=1$ (单位长度向量). 令 $H=I-2\mathbf{v}\cdot\mathbf{v}^*\in\mathbb{C}^{n\times n}$ 称为 Householder 矩阵.

（1）我们可以取

$$v=\frac{(1-\sqrt{3},1,1)^T}{\left\|(1-\sqrt{3},1,1)^T\right\|_2}.$$

即可有

$$H = I - 2vv^*, Hb = \begin{pmatrix} \sqrt{3} \\ 0 \\ 0 \end{pmatrix}.$$

备注：若目标是将实数向量a变成等 2-范数的实数向量b，可以取

$$v = \frac{a-b}{\|a-b\|_2}, H = I - 2vv^* = H = I - 2vv^T.$$

定义 5.5 (Givens 变换)

$\forall \theta \in \mathbb{R}$, 记 $s = \sin\theta$, $c = \cos\theta$, $J = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$ 是一个正交阵.

$\forall \mathbf{x} \in \mathbb{R}^2$, $J\mathbf{x}$ 表示将向量 $\mathbf{x}$ 旋转 θ 角 (弧度). 推广到 n 维情形, 令

$$J(j,k;\theta) = \begin{pmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & c & & & s & & \\ & & & & 1 & & & & \\ & & & & & \ddots & & & \\ & & & & & & 1 & & \\ & & & -s & & & & c & \\ & & & & & & & & 1 \\ & & & & & & & & & \ddots \\ & & & & & & & & & & 1 \end{pmatrix} \text{— Givens变换.}$$

（2）我们无法做到“一步到位”，但是可以分两步处理。

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \\ & & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \sqrt{2} \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} \frac{\sqrt{2}}{\sqrt{3}} & & \frac{1}{\sqrt{3}} \\ & 1 & \\ -\frac{1}{\sqrt{3}} & & \frac{\sqrt{2}}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \sqrt{3} \\ 0 \\ 0 \end{pmatrix}.$$